

NODE.JS

Building Your First CLI Tool [cite: 1, 3]

Your Command Center for School Tasks [cite: 4, 5]

Planrs Academy

Coding Adventures for Grade 6

What is a CLI? [cite: 6]

CLI stands for **Command Line Interface**. It's a way to talk to your computer by typing secret codes! [cite: 6, 13]

Why is it cool? [cite: 7-12]

- It's super **Fast!** [cite: 7]
- No Clicking—just typing! [cite: 9]
- It feels like casting **Magic Spells** [cite: 10]
- It gives you **Power** and control! [cite: 11, 12]

CLI vs GUI [cite: 14]

Most people use a **GUI** (Graphical User Interface). That's like a tablet screen with big, friendly buttons. [cite: 14, 16]

The Remote Analogy [cite: 15, 16]

A CLI is like a **Remote Control**. You don't have to walk up to the TV to press buttons. You just type the right "channel" and the computer does it instantly! [cite: 15, 16]

Developers love CLI because it lets them run magic spells to do many tasks quickly! [cite: 85]

Meet NPM! [cite: 18]

NPM is your **Magic Package Store**! It's full of "Coding Tools" that you can add to your project. [cite: 20, 23]

Popular Tools in the Store: [cite: 24-30]

- **chalk**: For neon colors! [cite: 24, 73]
- **figlet**: For giant text art! [cite: 30]
- **ora**: For cool loading spinners! [cite: 29]

Getting Started with npm init [cite: 34, 35]

Before we build, we need a "Project ID Card." We use a magic spell for this! [cite: 37, 39]

```
npm init -y [cite: 37]
```

This command creates a file called **package.json**. This file keeps track of all the magic tools we "buy" from the NPM store! [cite: 37]

Your Message Postman [cite: 42]

When you type a command, how does the code know what you said? Meet **process.argv**! [cite: 42]

Think of it as a **Postman**. It takes the message from your keyboard and delivers it to the Node.js "Brain." [cite: 42, 47-49]

It captures your **Input** and turns it into **Args** (Arguments) that the computer can read. [cite: 50, 51]

Creating Your First Script [cite: 52]

Let's make a script that says Hello when you tell it your name! [cite: 53, 55]

```
const name = process.argv[2]; [cite: 56]
console.log(`Hello, ${name}! Welcome to your Command Center.`);
[cite: 57]
```

Save this as **assistant.js** and get ready to run it! [cite: 58]

Running Your Script [cite: 59]

Open your terminal and type this command to see the magic: [cite: 61]

```
node assistant.js Arjun [cite: 62]
```

Output:

Hello, Arjun! Welcome to your Command Center. [cite: 63, 64]

You just told the computer your name using a secret command! [cite: 65, 66]

Building the Homework Helper [cite: 67, 68]

Now, let's make it more useful! We can capture your school tasks and print them out. [cite: 69]

```
const task = process.argv[2];  
console.log("Added to list: " + task); [cite: 70]
```

process.argv captures what you type after the command. **console.log** prints it back to you! [cite: 69, 70]

Adding Colors with Chalk [cite: 71]

The standard terminal text is a bit boring. Let's add some neon paint! [cite: 72, 73]

chalk is like neon paint for your terminal messages. It makes your CLI look cool and fun! [cite: 73-76]

Install it using this magic command: [cite: 77]

```
npm install chalk [cite: 78]
```

Using Chalk in Your Tool [cite: 80]

First, we need to import the package, then wrap our text in a color! [cite: 79]

```
const chalk = require('chalk'); [cite: 79]  
console.log(chalk.green("Task Added Successfully!")); [cite: 80]
```

Now your Homework Helper will talk to you in bright green! [cite: 80]

Fun CLI Ideas from India! [cite: 84]

Now that you know the basics, what else can you build? [cite: 87]

- **Pincode Finder:** Look up any Indian city! [cite: 81]
- **Cricket Score Notifier:** Live match updates! [cite: 81]
- **Vedic Maths Quiz:** Practice math tricks! [cite: 82]
- **Festival Countdown:** Days until Diwali or Holi! [cite: 83]
- **Tiffin Menu Planner:** Plan your lunch box! [cite: 83]

Why Developers Love CLI [cite: 85]

Typing commands isn't just for fun—it's like having a superpower! [cite: 86]

It lets you automate boring tasks and control your computer exactly how you want. [cite: 85]

Every great programmer starts by mastering their "Command Center." [cite: 87]

Your Adventure Begins [cite: 87]

Building a CLI tool is the first step toward becoming a full-stack developer.
[cite: 87]

Imagine: Your code running on servers all around the world!

Keep coding, keep exploring, and never stop building! [cite: 87]

THANK YOU! [cite: 87]

Any Questions? [cite: 87]

Ready to build your own CLI tool? [cite: 87]

Adventure awaits at your Command Center! [cite: 87]